ADICHUNCHANAGIRI POLYTECHNIC

(Affiliated to the Board of Technical Examinations, Karnataka)

Department of Mechanical Engineering

PROGRAMMING CONCEPTS IN MECHANICAL ENGINEERING

LABORATORY RECORD MANUAL

Student Name : _____

USN / Register Number : _____

Semester / Branch : III Semester, Diploma in Mechanical Engineering

Batch / Section : _____

Academic Year : _____

CERTIFICATE

This is to certify that this is a bona fide record of practical work done in the Programming Concepts in Mechanical Engineering laboratory by the student named above, of the III Semester Diploma in Mechanical Engineering, during the academic year mentioned above, at Adichunchanagiri Polytechnic.

Signature of Student | Signature of Staff-in-Charge | Signature of HOD

INSTRUCTIONS TO STUDENTS

1. Write the Aim, Algorithm/Formula and Program for each experiment neatly in this manual before coming to the laboratory.
2. Execute each program on the computer as per the schedule given by the staff-in-charge.
3. After successful execution, write down the actual output obtained in the ruled space provided, by hand.
4. Get the output verified and signed by the staff-in-charge on the same day of execution.
5. Do not copy output from another student; record only the output obtained from your own execution.
6. Keep this manual neat and bring it to every laboratory session along with a pen/pencil.

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Week 1: Algorithms And Flowcharts

Ex. No: 1 Date: _____

1. Convert Fahrenheit to Centigrade

Aim: Write the algorithm and flowchart to convert a given Fahrenheit temperature to Centigrade.

Formula: C = 5(F - 32) / 9

Algorithm:

1. Start
2. Read Fahrenheit temperature F
3. Calculate C = (5/9) x (F - 32)
4. Display Centigrade temperature C
5. Stop

Flowchart: Start -> Input Fahrenheit temperature F -> Process: C = 5/9 x (F-32) -> Display Centigrade temperature C -> Stop

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 2 Date: _____

2. Efficiency of a Simple Machine

Aim: Write the algorithm and flowchart to calculate the efficiency of a simple machine.

Formula: eta = (Output Work / Input Work) x 100

Algorithm:

1. Start
2. Read x and y
3. Calculate Efficiency = (y / x) x 100
4. Display Efficiency
5. Stop

Note: x represents Input Work, y represents Output Work. Flowchart: Start -> Input x, y -> Process: Efficiency = (y/x) x 100 -> Display Efficiency -> Stop

Output:

Result:

_____ Verified by (Staff Signature): Date:

Ex. No: 3 Date: _____

3. Generate the Multiplication Table for the Number 5 (Control Loop)

Aim: Write the algorithm and flowchart to generate the multiplication table for the number 5.

Algorithm:

1. Start
2. Set i = 1
3. Repeat while i <= 10
4. Print 5 x i
5. Increment i by 1
6. Stop

Flowchart: Start -> i=1 -> Is i<=10? -> Yes: Print 5xi, i=i+1 (loop back) -> No: Stop

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 4 Date: _____

4. Finding the Max of Three Forces (Decision)

Aim: Write the algorithm and flowchart to find the maximum of three forces F1, F2, F3.

Algorithm:

1. Start
2. Read F1, F2, F3
3. If F1 > F2 and F1 > F3, then Max = F1
4. Else if F2 > F1 and F2 > F3, then Max = F2
5. Else Max = F3
6. Display Max Force
7. Stop

Flowchart: Start -> Input F1,F2,F3 -> Is F1>F2 and F1>F3? -> Yes: Display Max=F1 -> No: Is F2>F1 and F2>F3? -> Yes: Display Max=F2 -> No: Display Max=F3 -> Stop

Output:

Result:

_____ Verified by (Staff Signature): Date:

Week 2: Introduction To Python And Basic Constructs

Ex. No: 5 Date: _____

1. Accept User Input and Perform Arithmetic Operations

Aim: Write a Python program to accept user input and perform arithmetic operations (addition, subtraction, etc.)

Program:

```
a = int(input("Enter first number : "))
b = int(input("Enter second number : "))
print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 6 Date: _____

2. Calculate Output Speed and Torque in a Gear System

Aim: Calculate the output speed and torque in a gear system based on user-inputted values for input speed, input torque and gear ratio using basic operators.

Program:

```
input_speed = float(input("Enter Input Speed : "))
input_torque = float(input("Enter Input Torque : "))
gear_ratio = float(input("Enter Gear Ratio : "))
output_speed = input_speed / gear_ratio
output_torque = input_torque * gear_ratio
print("Output Speed =", output_speed)
print("Output Torque =", output_torque)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 7 Date: _____

3. Practice Problems on Type Conversions and Working with Variables

Aim: Practice problems on type conversions and working with variables.

Program:

```
a = int(input("Enter first number : "))
b = int(input("Enter second number : "))
sum = a + b
print("Addition =", sum)
average = (a + b) / 2
print("Average =", average)
x = float(a)
print("Float Value =", x)
y = str(b)
print("String Value =", y)
```

Output:

Result:

_____ Verified by (Staff Signature): Date:

Week 3: Control Structures In Python

Ex. No: 8 Date: _____

1. Determine if a Number is Even or Odd

Aim: Write a Python program to determine if a number is even or odd.

Program:

```
num = int(input("Enter a number : "))
if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 9 Date: _____

2. Print Multiplication Tables Using a For Loop

Aim: Create a program that uses a for loop to print multiplication tables.

Program:

```
n = int(input("Enter a number : "))
for i in range(1, 11):
    print(n, "x", i, "=", n*i)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 10 Date: _____

3. Calculate the Factorial of a Number Using a While Loop

Aim: Write a Python program to calculate the factorial of a number using a while loop.

Program:

```
n = int(input("Enter a number : "))
fact = 1
i = 1
while i <= n:
    fact = fact * i
    i = i + 1
print("Factorial =", fact)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 11 Date: _____

4. Practice Program Using If-else and Loops

Aim: Practice program using if-else and loops. (a) Check positive/negative number, (b) Sum of first N natural numbers using for loop, (c) Reverse a number using while loop.

Program:

```
# (a) Check positive or negative number
n = int(input("Enter a number : "))
if n > 0:
    print("Positive Number")
else:
    print("Negative Number")

# (b) Sum of first N natural numbers using for loop
n = int(input("Enter a number : "))
sum = 0
for i in range(1, n+1):
    sum = sum + i
print("Sum =", sum)

# (c) Reverse a number using while loop
n = int(input("Enter a number : "))
reverse = 0
while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n = n // 10
print("Reverse Number =", reverse)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 4: Functions And Modules

Ex. No: 12 Date: _____

1. Function to Calculate the Area of a Circle

Aim: Write a Python program that defines a function to calculate the area of a circle.

Program:

```
def area(r):
    a = 3.14 * r * r
    print("Area =", a)

radius = float(input("Enter radius : "))
area(radius)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 13 Date: _____

2. Find the GCD of Two Numbers Using a Function

Aim: Create a program to find the greatest common divisor (GCD) of two numbers using a function.

Program:

```
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

x = int(input("Enter first number : "))
y = int(input("Enter second number : "))
print("GCD =", gcd(x, y))
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 14 Date: _____

3. Explore Lambda Functions

Aim: Explore lambda functions by creating small one-liners for tasks like squaring a number.

Program:

```
square = lambda x : x * x
print(square(5))
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 15 Date: _____

4. Use the Math Module for Trigonometric Calculations

Aim: Write a Python program to use the math module for trigonometric calculations.

Program:

```
import math
x = math.sin(0)
y = math.cos(0)
print("Sin Value =", x)
print("Cos Value =", y)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 5: Data Structures In Python

Ex. No: 16 Date: _____

1. Store Names of Students in a List and Perform Operations

Aim: Create a Python program that stores names of students in a list and performs operations like adding, removing and sorting.

Program:

```
students = ["Ravi", "Kiran", "Asha"]
students.append("Manu")
print(students)
students.remove("Kiran")
print(students)
students.sort()
print(students)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 17 Date: _____

2. Convert a List of Temperatures from Celsius to Fahrenheit

Aim: Write a Python program that converts a list of temperatures from Celsius to Fahrenheit.

Program:

```
celsius = [0, 20, 30, 40]
for c in celsius:
    f = (9/5) * c + 32
    print(f)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 18 Date: _____

3. Dictionary for Storing Student Records

Aim: Create a Python dictionary for storing student record (name, age and grade) and write functions to add and remove entries.

Program:

```
student = {
    "name" : "Ravi",
    "age" : 18,
    "grade" : "A"
}
student["branch"] = "Mechanical"
print(student)
del student["age"]
print(student)
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 19 Date: _____

4. Use Tuples to Store and Display Coordinates in 2D Space

Aim: Write a program that uses tuples to store and display coordinates in 2D space.

Program:

```
point = (10, 20)
print("X Coordinate =", point[0])
print("Y Coordinate =", point[1])
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 6: File Handling And Exception Handling

Ex. No: 20 Date: _____

1. Read from a File and Count the Number of Words

Aim: Write a Python program to read from a file and count the number of words in it.

Program:

```
file = open("sample.txt", "r")
data = file.read()
words = data.split()
print("Number of Words =", len(words))
file.close()
```

Note: sample.txt file must be stored in the same folder where the Python program is saved. (Content for sample.txt: 'Python is easy to learn').

Output:

Result:

_____ Verified by (Staff Signature): Date:

Ex. No: 21 Date: _____

2. Write a List of Random Numbers to a File and Read Them Back

Aim: Create a Python program that writes a list of random numbers to a file and reads them back.

Program:

```
import random
file = open("numbers.txt", "w")
for i in range(5):
    number = random.randint(1, 100)
    file.write(str(number) + "\n")
file.close()
file = open("numbers.txt", "r")
print(file.read())
file.close()
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 22 Date: _____

3. Handle Division by Zero Using try-except Blocks

Aim: Write a program to handle division by zero using try-except blocks.

Program:

```
try :
    a = 10 / 0
except ZeroDivisionError :
    print("Cannot divide by zero")
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 23 Date: _____

4. Read Data from a CSV File and Perform Basic Operations

Aim: Write a Python program to read data from a CSV file and perform basic operations like counting the number of rows.

Program:

```
import csv
file = open("student.csv", "r")
data = csv.reader(file)
count = 0
for row in data :
    count = count + 1
print("Number of Rows =", count)
file.close()
```

Note: student.csv file must be stored in the same folder where the Python program is saved. (Content: Name,Age / Ravi,18 / Manu,19 / Asha,17).

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 7: Object-Oriented Programming (Oop)

Ex. No: 24 Date: _____

1. Car Class with Attributes and Methods

Aim: Write a Python class to represent a Car with attributes like model, year and color. Include methods to start, stop and display car details.

Program:

```
class Car :
    def __init__(self, model, year, color) :
        self.model = model
        self.year = year
        self.color = color
    def start(self) :
        print("Car Started")
    def stop(self) :
        print("Car Stopped")
    def display(self) :
        print("Model =", self.model)
        print("Year =", self.year)
        print("Color =", self.color)

c = Car("Toyota", 2024, "Red")
c.start()
c.display()
c.stop()
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 25 Date: _____

2. Inheritance with Base Class Animal and Derived Classes Dog and Cat

Aim: Create a program that implements inheritance with a base class Animal and derived classes Dog and Cat.

Program:

```
class Animal :
    def sound(self) :
        print("Animal makes sound")

class Dog(Animal) :
    def bark(self) :
        print("Dog barks")

class Cat(Animal) :
    def meow(self) :
        print("Cat meows")

d = Dog()
d.sound()
d.bark()
c = Cat()
c.sound()
c.meow()
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 26 Date: _____

3. Demonstrate Method Overriding

Aim: Write a Python program that demonstrates method overriding.

Program:

```
class Animal :
    def sound(self) :
        print("Animal makes sound")

class Dog(Animal) :
    def sound(self) :
        print("Dog barks")

d = Dog()
d.sound()
```

Output:

Result:

_____ Verified by (Staff Signature): Date:

Ex. No: 27 Date: _____

4. Bank Class Supporting Deposit, Withdrawal and Balance Inquiry

Aim: Develop a Python class for a bank account that supports deposit, withdrawal and balance inquiry.

Program:

```
class Bank :
    def __init__(self) :
        self.balance = 1000
    def deposit(self) :
        self.balance = self.balance + 500
    def withdraw(self) :
        self.balance = self.balance - 200
    def display(self) :
        print("Balance =", self.balance)

b = Bank()
b.deposit()
b.display()
b.withdraw()
b.display()
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 8: Introduction To Raspberry Pi And Setup

Ex. No: 28 Date: _____

1. Controlling DC Motors Using L298N Motor Driver with Raspberry Pi

Aim: Write a program to control DC motors using L298N motor driver with Raspberry Pi.

Program:

```
from gpiozero import Motor
from time import sleep
motor = Motor(forward=17, backward=18)
motor.forward()
print("Motor Rotating Forward")
sleep(5)
motor.stop()
print("Motor Stopped")
```

Note: Before executing this program, install the gpiozero library using: pip install gpiozero. Requires Raspberry Pi hardware with L298N motor driver connected as per Fig. 8.5.

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 29 Date: _____

2. Building a Simple Automated System (Fan / Water Pump Controller)

Aim: Build simple automated systems like a fan controller and a water pump controller.

Program:

```
# Program 1: Fan Controller
from gpiozero import Motor
temperature = 35
fan = Motor(forward=17, backward=18)
if temperature > 30 :
    fan.forward()
    print("Fan ON")
else :
    fan.stop()
    print("Fan OFF")

# Program 2: Water Pump Controller
water_level = 20
if water_level < 30 :
    print("Water Pump ON")
else :
    print("Water Pump OFF")
```

Note: Before executing GPIO programs, install the Raspberry Pi GPIO library using: pip install gpiozero. Requires Raspberry Pi hardware set up as per Fig. 8.6 and Fig. 8.7.

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 9: Building A Simple Robot Using Raspberry Pi

Ex. No: 30 Date: _____

1. Building a Simple Robot Using Raspberry Pi, Motors and Sensors

Aim: Build a simple robot using Raspberry Pi, motors and sensors.

Program:

```
from gpiozero import Motor
from time import sleep
left_motor = Motor(forward=17, backward=18)
right_motor = Motor(forward=22, backward=23)
left_motor.forward()
right_motor.forward()
print("Robot Moving Forward")
sleep(5)
left_motor.stop()
right_motor.stop()
print("Robot Stopped")
```

Note: Before executing sensor interfacing programs, install the required Python libraries using: pip install Adafruit_DHT, pip install Adafruit_BMP, pip install gpiozero. Requires Raspberry Pi hardware set up as per Fig. 9.7.

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 10: Statistical Quality Control (Sqc)

Ex. No: 31 Date: _____

1. Sample Average (X-bar) and Sample Range (R) Charts Using Matplotlib

Aim: Develop a Python program to calculate the Sample Average (X-bar) and Sample Range (R) for the given data and plot X-bar and R charts using Matplotlib.

Program:

```
import matplotlib.pyplot as plt
samples = [
    [10, 11, 12, 12, 11],
    [11, 12, 13, 12, 12],
    [11, 12, 13, 14, 12],
    [12, 13, 14, 13, 13],
    [11, 12, 13, 12, 11],
]
sample_no = []
xbar = []
r = []
for sample in samples :
    average = sum(sample) / len(sample)
    range_value = max(sample) - min(sample)
    sample_no.append(sample)
    xbar.append(average)
    r.append(range_value)
    print("Sample =", sample)
    print("Average =", average)
    print("Range =", range_value)

plt.plot(sample_no, xbar, marker='o')
plt.title("X Bar Chart")
plt.xlabel("Sample Number")
plt.ylabel("Average")
plt.show()

plt.plot(sample_no, r, marker='o')
plt.title("R Chart")
plt.xlabel("Sample Number")
plt.ylabel("Range")
plt.show()
```

Note: Before executing, install Matplotlib using: pip install matplotlib. The program produces two graphs (X Bar Chart and R Chart) - sketch/paste them in the output space below.

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 11: Introduction To Statistical Process Control (Spc)

Ex. No: 32 Date: _____

1. Calculate Mean, Variance and Standard Deviation

Aim: Calculate mean, variance and standard deviation using Python.

Program:

```
import statistics
data = [10, 12, 14, 16, 18]
mean = statistics.mean(data)
variance = statistics.variance(data)
std = statistics.stdev(data)
print("Mean =", mean)
print("Variance =", variance)
print("Standard Deviation =", round(std, 2))
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 33 Date: _____

2. Generate X-bar and R Control Charts; Identify In-control / Out-of-control Points

Aim: Generate X-bar and R control charts from sample data using Matplotlib, and identify in-control and out-of-control points.

Program:

```
import matplotlib.pyplot as plt
xbar = [11, 12, 13, 12, 11.5]
r = [2, 2, 2, 2, 2]
sample_no = [1, 2, 3, 4, 5]
plt.plot(sample_no, xbar, marker='o')
plt.title("X Bar Chart")
plt.xlabel("Sample Number")
plt.ylabel("Average")
plt.show()

plt.plot(sample_no, r, marker='o')
plt.title("R Chart")
plt.xlabel("Sample Number")
plt.ylabel("Range")
plt.show()

# Identify in-control and out-of-control points
data = [12, 13, 11, 15, 18, 10]
ucl = 16
lcl = 11
for value in data :
    if value > ucl or value < lcl :
        print(value, "- Out of Control")
    else :
        print(value, "- In Control")
```

Note: Before executing, install Matplotlib using: pip install matplotlib.

Output:

Result:

_____ Verified by (Staff Signature): Date:

Week 12: Hypothesis Testing And Total Quality Management

Ex. No: 34 Date: _____

1. Plot a Fishbone (Cause and Effect) Diagram

Aim: Generate Python code to plot a Fishbone diagram for the given data (causes: Machine, Material, Manpower, Method affecting a Defect).

Program:

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8,4))
plt.plot((10,10), (5,1))
plt.plot((12,4), (6,5))
plt.plot((12,4), (4,5))
plt.text(10,10,"Defect")
plt.text(1.2,4,"Machine")
plt.text(7.2,6,"Material")
plt.text(7.2,4,"Method")
plt.text(1.2,6,"Manpower")
plt.axis('off')
plt.title("Fishbone Diagram")
plt.show()
```

Output:

Result:

_____ Verified by (Staff Signature): Date:

Ex. No: 35 Date: _____

2. Plot Quality Control Chart Tools for Given Data

Aim: Generate Python code to plot different quality control tools for the given data.

Program:

```
import matplotlib.pyplot as plt
samples = [1, 2, 3, 4, 5]
values = [10, 15, 12, 18, 14]
plt.plot(samples, values, marker='o')
plt.title("Quality Control Chart")
plt.xlabel("Sample Number")
plt.ylabel("Quality Value")
plt.show()
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Week 13: Finite Element Method (Fem)

Ex. No: 36 Date: _____

1. Plot Stress Distribution in a Plate at Various Lengths Along the Bar

Aim: Write a Python code to plot the stress distribution in the plate at various lengths along the bar.

Program:

```
import matplotlib.pyplot as plt
length = [0, 1, 2, 3, 4, 5]
stress = [10, 15, 20, 18, 14, 12]
plt.plot(length, stress, marker='o')
plt.title("Stress Distribution Along a Bar")
plt.xlabel("Length of Bar")
plt.ylabel("Stress")
plt.grid(True)
plt.show()
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____

Ex. No: 37 Date: _____

2. Compare and Plot Stress Distribution in a Plate With and Without Hole

Aim: Write a Python code to compare and plot the stress distribution in the plate with and without hole.

Program:

```
import matplotlib.pyplot as plt
length = [0, 1, 2, 3, 4, 5]
stress_without_hole = [10, 12, 14, 13, 12, 11]
stress_with_hole = [10, 18, 25, 22, 16, 12]
plt.plot(length, stress_with_hole, marker='o', label="With Hole")
plt.plot(length, stress_without_hole, marker='o', label="Without Hole")
plt.title("Stress Distribution Comparison")
plt.xlabel("Length of Plate")
plt.ylabel("Stress")
plt.grid(True)
plt.legend()
plt.show()
```

Output:

Result:

_____ Verified by (Staff Signature): Date: _____